

The sharpest way to provide PROTECTION against infection

Vaccination

Vaccination exposes a person to a mild form of a disease, so that his or her body learns to fight the disease in the future.

This pin device was used in 18th-century Europe to move infected tissue from one person to another.



Variolation

In the 10th century, the Chinese began a simple form of vaccination called **VARIOLATION**. They exposed healthy people to the disease smallpox, in the hope that they would catch a milder version and, after recovery, *become immune*. A piece of a scab would be placed under the skin, or ground up and blown up a patient's nose. The practice spread to Europe, but the method was unreliable and could lead to the **spread of other diseases**.



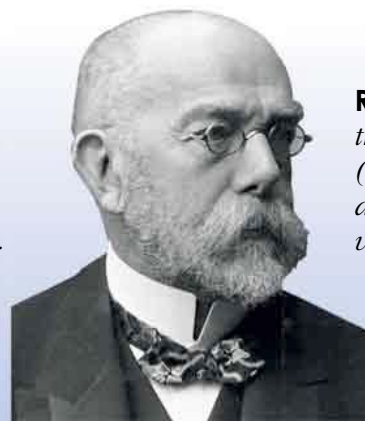
Dairy cows carried the cowpox virus on their udders.

Milkmaids often caught cowpox, a disease caused by a virus related to but much milder than deadly smallpox.

It paved the way for...



In 1853, French doctor **Charles Pravaz** invented the practical metal **HYPODERMIC SYRINGE**. Used to administer vaccines, it uses a hollow needle to pierce the skin.



ROBERT KOCH discovered the bacteria that cause **ANTHRAX** (1876), **TUBERCULOSIS** (1882), and **CHOLERA** (1883), allowing vaccines to be developed.